

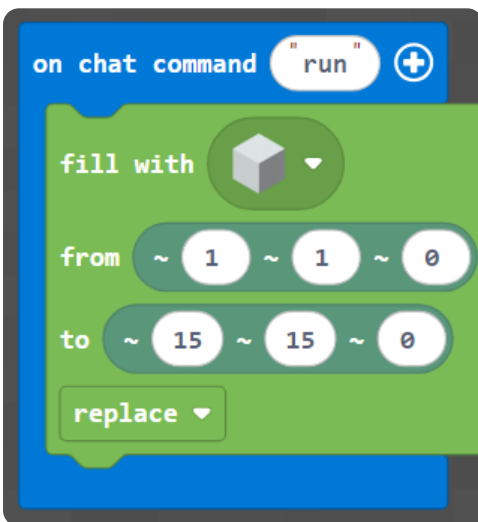
## M003 Week 4 — English Worksheet (Advanced)

**Topic:** Pixel Art 2 — Cat & Tree (x, y) · **Course:** 3D Coordinates · **Level:** Advanced · **Time:** about 45 minutes

This week you copy **two pictures** onto your wall — a **cat face** and a **tree**. Same big idea as last week: every block sits at a spot named by **two numbers** — (x, y). x is how far **across**, y is how far **up**. Read the picture, work out each block's (x, y), and place it. The tricky part this week is reading shapes that lean and spread, so think carefully before you build.

**First, build your canvas.** Run this to make a blank 15 × 15 wall, then put a **red block at your feet** as your **home spot**:

### Blocks



### Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Big idea:** (x, y) is counted from your **home spot** — not from the corner of the world. Stand on the red block *every* time you run, or the whole picture shifts. Move your feet, move your picture.

### 1 · Predict 🧠

Read each set of steps. For each one, write what you think you will see **and the reason** you think so. Do not run it yet.

```
place brown block at (8, 1)
place brown block at (8, 2)
place brown block at (8, 3)
place brown block at (8, 4)
```

**What shape is this, across or up? Which number stays the same, and why does that make the line go that way?**

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```
place orange block at (2, 15)
place orange block at (3, 15)
place orange block at (13, 15)
place orange block at (14, 15)
```

**Two pairs of blocks, both at  $y = 15$  but far apart. What part of the cat could each pair be, and why are they in two separate groups?**

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```
place green block at (8, 15)
place green block at (7, 14)
place green block at (9, 14)
```

**These green blocks sit high and spread out as they go down. Do they make a fruit or the top of the tree, and what is happening to  $x$  and  $y$  at the same time?**

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```
place green block at (1, 9)
place green block at (1, 10)
place brown block at (8, 1)
place brown block at (8, 2)
```

Two different colours in one run. Describe both lines: which is leaves, which is trunk, and how can you tell from the numbers alone?

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## 2 · Spot the Bug 🐛

Each block of code below is broken. Read what it is **supposed** to do, rewrite it so it works, then explain in 2 or 3 sentences why the original was wrong and why your fix works.

**Bug A** — This should make the **tree trunk**: four brown blocks going **up** at  $x = 8$ , from (8, 1) to (8, 4). Right now they all land in the **same spot**.

```
place brown block at (8, 1)
place brown block at (8, 1)
place brown block at (8, 1)
place brown block at (8, 1)
```

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug B** — The tree trunk should grow **up** at  $x = 8$ : (8, 2), (8, 3), (8, 4). Right now it grows **across** instead.

```
place brown block at (2, 8)
place brown block at (3, 8)
place brown block at (4, 8)
```

**Write the fixed code:**

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Why was it wrong? Why does your fix work?

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**Bug C (think hard — the numbers are not the problem).** A student typed every coordinate of the cat correctly, but the whole cat came out **one block too far to the right and one block too low** every single time. The code is fine. Something about *how* they ran it was wrong.

```
place orange block at (2, 15)
place orange block at (3, 15)
place orange block at (13, 15)
place orange block at (14, 15)
```

What did the student do wrong, and what should they do instead? Explain why the picture shifted even though the numbers were right.

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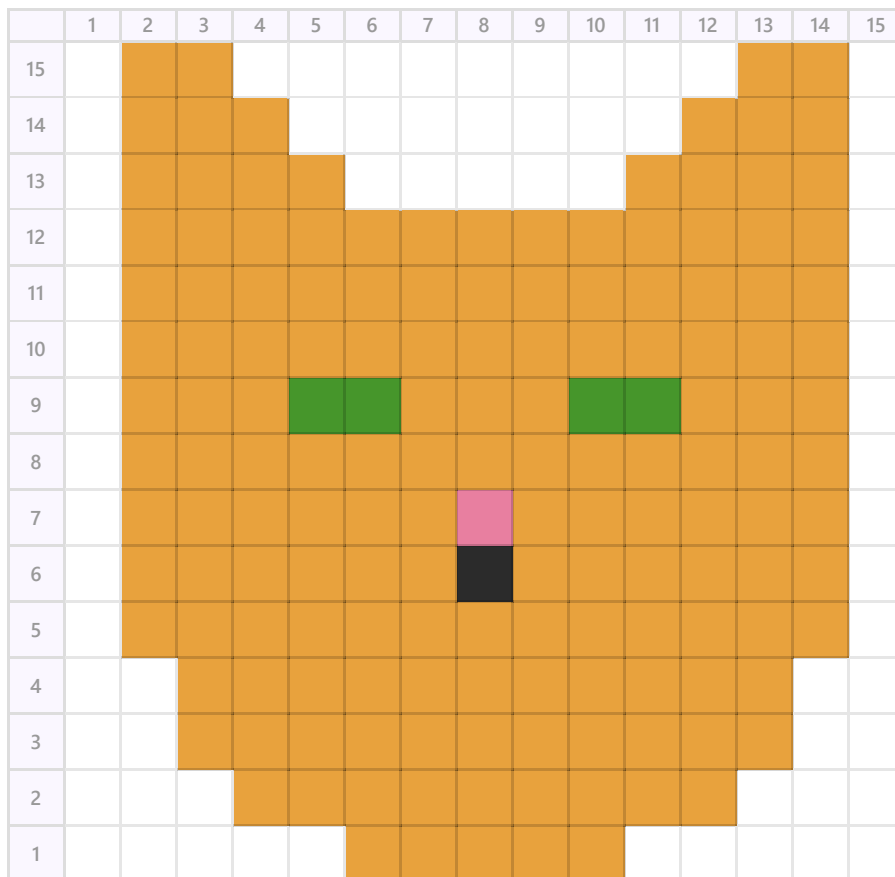
### 3 · Build It 📷

Switch to your homework world and stand on your **home spot** (the red block). Plan first, then build. The numbers across the **top** are **x**, the numbers down the **side** are **y**.

**Plan your coordinates first.** Before you place anything, write the (x, y) of the **lowest block** and the **highest block** of the picture you will build. This stops the picture running off the top of the wall.

Now build the 🐱 cat **and** the 🌳 tree below by reading the (x, y) of every colored square.

 **Cat face**



## Tree

|    | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|----|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|
| 15 |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
| 14 |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
| 13 |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
| 12 |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
| 11 |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
| 10 |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
| 9  |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
| 8  |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
| 7  |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
| 6  |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
| 5  |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
| 4  |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
| 3  |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
| 2  |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |
| 1  |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |

**Modify challenge.** Once both pictures are built, change one thing with your own idea — give the cat a different eye colour, add an apple to the tree, or make the trunk taller. Build your change, and be ready to say which coordinates you added or swapped.

When everything is finished, send a photo (or a short video). Then explain what you did with these sentence starters — write 5 or 6 sentences total.

First, I planned the coordinates by ...

The lowest block in my picture is at (x = ..., y = ...) and the highest is at (x = ..., y = ...).

To go across a row I changed the ... number; to go up I changed the ... number.

The trickiest part to read off the grid was ... because ...

For my modify challenge, I changed ... by placing blocks at ...

If I built this again, I would ...

First, I planned the coordinates by ...

The lowest block in my picture is at (x = ..., y = ...) and the highest is at (x = ..., y = ...).

To go across a row I changed the ... number; to go up I changed the ... number.

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The trickiest part to read off the grid was ... because ...

For my modify challenge, I changed ... by placing blocks at ...

If I built this again, I would ...



#### 4 · Explain It

Take a video on a phone while you show both pictures on the wall. Talk through it like you are teaching someone who has never seen it. Try to use these words: **x, y, coordinate, across, up, home spot**.

1. Show your home spot (the red block) and say in your own words why standing on it matters.
2. Show your finished cat and tree from the front.
3. Point at one block and read its coordinate out loud: "this one is at x ..., y ...".
4. Show your modify change and explain which coordinates you added or swapped.

**Write a short paragraph (5 or 6 sentences) saying what you will say in your video. Then answer: if you were teaching a friend who has never done this, what would you teach them first, and why?**

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**Submit** 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.